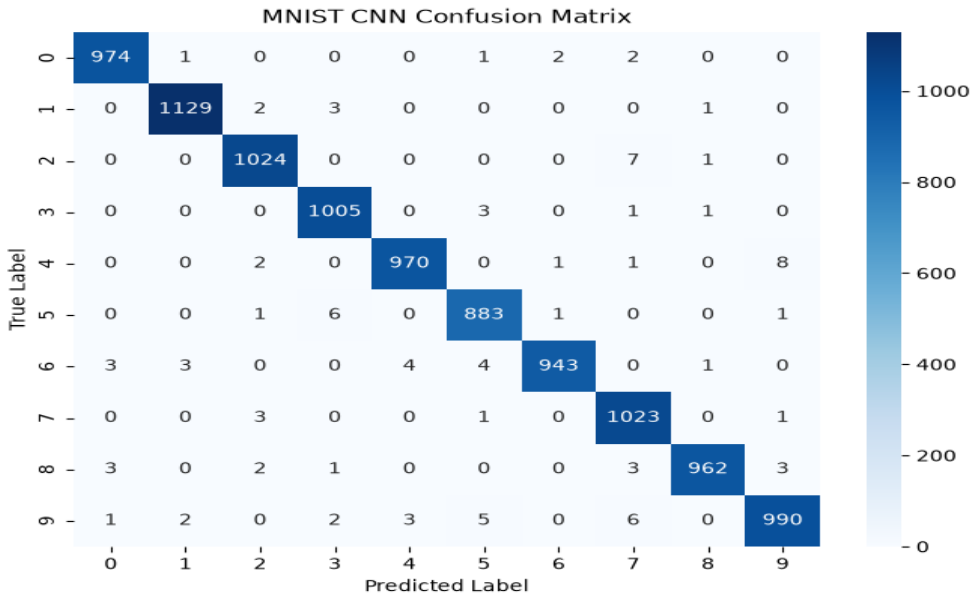
Lab Sheet-10

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```
import os
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras import layers, models
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
# 1. Dataset Load & Preprocessing (MNIST Digit Recognition)
(X_train, y_train), (X_test, y_test) = tf.keras.datasets.mnist.load_data()
# Normalize pixel values (0-255 to 0-1) and reshape for CNN input
X_train = X_train.astype('float32') / 255.0
X_test = X_test.astype('float32') / 255.0
X_train = np.expand_dims(X_train, axis=-1)
X_test = np.expand_dims(X_test, axis=-1)
# 2. CNN Model Architecture Design
model = models.Sequential([
  layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
  layers.MaxPooling2D((2, 2)),
  layers.Conv2D(64, (3, 3), activation='relu'),
  layers.MaxPooling2D((2, 2)),
  layers.Flatten(),
  layers.Dense(64, activation='relu'),
  layers.Dense(10, activation='softmax')
])
# 3. Model Configuration
model.compile(optimizer='adam',
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy'])
# 4. Model Training
history = model.fit(X_train, y_train, epochs=5, batch_size=64, validation_split=0.1)
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# 5. Model Evaluation
test_loss, test_acc = model.evaluate(X_test, y_test, verbose=0)
print(f"\nTest Accuracy: {test_acc * 100:.2f}%")
# 6. Predictions & Performance Metrics
y_pred = model.predict(X_test)
y_pred_classes = np.argmax(y_pred, axis=1)
print("\n--- Classification Report ---")
print(classification_report(y_test, y_pred_classes))
# 7. Plot & Save Confusion Matrix in current folder
cm = confusion_matrix(y_test, y_pred_classes)
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title('MNIST CNN Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.savefig('experiment10/confusion_matrix.png')
plt.close()
print("\nExecution complete! Saved 'confusion_matrix.png' in experiment10 folder.")
```



Questions & Answers

Q1. What is image classification? How does it differ from object detection and image segmentation?

- **Image Classification:** Assigns a single categorical label to an entire image without locating spatial boundaries.
- **Object Detection:** Identifies multiple objects, assigns labels, and draws bounding boxes around their locations.
- **Image Segmentation:** Classifies every pixel to outline exact visual boundaries of objects.

Q2. Explain the architecture and working principle of a CNN.

A Convolutional Neural Network (CNN) consists of convolutional layers, pooling layers, activation functions, and fully connected layers. It automatically learns feature hierarchies (edges -> textures -> shapes) by sliding spatial filters (kernels) across input matrices.

Q3. What is the role of Convolutional Layers, Pooling Layers, and Fully Connected Layers in a CNN?

- **Convolutional Layer:** Extracts spatial feature maps using learnable filters.
- **Pooling Layer:** Reduces spatial dimensions (downsampling) to lower computation and prevent overfitting.
- **Fully Connected (Dense) Layer:** Combines extracted features to output final class probability distributions.

Q4. Why is image normalization performed before training a deep learning model?

Normalization scales pixel values (e.g., $0 - 255$to $0 - 1$). It stabilizes gradient descent, speeds up optimization convergence, and prevents exploding/vanishing gradient issues.

Q5. Explain the purpose of activation functions such as ReLU and Softmax in CNNs.

- **ReLU ($\max(0, x)$):** Introduces non-linearity, aiding the learning of complex relationships while solving vanishing gradient problems.
- **Softmax:** Converts final output logits into normalized probability distributions that sum up to 1 across target classes.

Q6. What is a Confusion Matrix? How is it used to evaluate classification performance?

A Confusion Matrix is an $N \times N$layout comparing Ground Truth classes against Predicted classes. It breaks down classification into True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Q7. Differentiate between Accuracy, Precision, Recall, and F1-Score with suitable examples.

- **Accuracy:** $\frac{TP+TN}{\text{Total}}$(Overall correctness ratio).
- **Precision:** $\frac{TP}{TP+FP}$(Exactness of positive predictions).

- **Recall:** $\frac{TP}{TP+FN}$(Completeness in detecting actual positives).
- **F1-Score:** $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}}$(Harmonic mean balancing Precision and Recall).

Q8. Compare traditional feature-based image classification methods with CNN-based classification.

Feature	Traditional Methods (e.g., SIFT/HOG + SVM)	CNN-Based Classification
Extraction	Requires handcrafted features	Fully automated feature extraction
Scalability	Struggles on complex/unstructured data	Scales well with big data and deep networks
Pipeline	Separate extraction and classification steps	End-to-end learning pipeline

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Q9. Mention five real-world applications of CNN-based image classification.

1. Medical Diagnostics (X-ray/MRI scan disease identification)
2. Biometric Authentication (facial recognition systems)
3. Automated Document Processing (OCR digit/character recognition)
4. Quality Inspection (assembly line defect detection)
5. Autonomous Driving (traffic sign categorization)

Q10. How can techniques such as Data Augmentation, Transfer Learning, and Hyperparameter Tuning improve performance?

- **Data Augmentation:** Increases dataset diversity using random rotations/flips, mitigating overfitting.
- **Transfer Learning:** Fine-tunes pre-trained models on smaller target datasets to boost accuracy.
- **Hyperparameter Tuning:** Optimizes learning rate, batch size, and network depth to improve model stability and accuracy.